COMPLETELY DECOMPOSED MORPHISMS AND COVERINGS ARE STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 23165-23173 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 23175-23184, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Companion to the collected decomposition card L16068 (decompose-completely-decomposed); base-change stability is the additional aggregate.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definition of completely decomposed morphism/family (chapter definition section: every point of the base has exactly one preimage with trivial residue extension); Schemes, lemma points-fibre-product (residue field computation of fibre product points).

Source reference. Original lemma 01JT: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23184–23184. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The base change of a completely decomposed morphism of schemes is completely decomposed. If $\{f_i : X_i \to Y\}_{i \in I}$ is completely decomposed and $Y' \to Y$ is a morphism of schemes, then $\{X_i \times_Y Y' \to Y'\}_{i \in I}$ is completely decomposed.*

Proof. Let $f : X \to Y$ and $g : Y' \to Y$ be morphisms of schemes. Let $y' \in Y'$ be a point with image $y = g(y')$ in Y. If $x \in X$ is a point such that $f(x) = y$ and $\kappa(x) = \kappa(y)$, then there exists a unique point $x' \in X' = X \times_Y Y'$ which maps to y' in Y' and to x in X and moreover $\kappa(x') = \kappa(y')$, see Schemes, Lemma 01JT. From this fact the lemma follows easily; we omit the details. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.